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Curtailed electricity and hydrogen in Association of Southeast Asian Nations and East Asia Summit: Perspectives form an economic and environmental analysis

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HIGHLIGHTS

- A cost-benefit analysis is performed for producing hydrogen from curtailed electricity.
- Hydrogen from curtailed electricity appears not to be cost-effective with current carbon prices.
- Carbon price about $15 per tonne of CO_2 justifies hydrogen from renewables for both ASEAN and EAS.
- High electrolyser load factors make hydrogen from renewables cost-competitive.
- Low electrolyser load factors require significantly high carbon prices.

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ABSTRACT

This study examines how well producing hydrogen via electrolysis from curtailed electricity from renewables could fulfil environmental benefits against the cost of producing hydrogen via electrolysis in the context of the Association of Southeast Asian Nations (ASEAN) and the East Asia Summit (EAS). The cost of producing hydrogen via electrolysis ranges from less than USD[2] per kgH_2 when the electrolyser load factor is 1500 h or above to USD10 per kgH_2 or even higher when the electrolyser load factor is 500 h or lower. The amount of CO_2 emissions abated by hydrogen produced from curtailed electricity from renewables ranges from about 130 million tonnes to about 150 million tonnes for ASEAN and from about 18,000 million tonnes to about 19,000 million tonnes for EAS. Applying prevailing carbon prices to the CO_2 emissions abated, the possible monetised benefits of hydrogen produced via electrolysis from curtailed electricity from renewables range from about USD0.25 per kgH_2 to about USD9.00 per kg H_2 for ASEAN and from about USD0.50 per kgH_2 to about USD15.00 per kg H_2 for EAS. The results of the cost-benefit analysis suggest that the price of carbon needs to be about USD15 per tonne of CO_2 to justify hydrogen produced via electrolysis from curtailed electricity from renewables for both ASEAN and EAS. The results also suggest that high electrolyser load factors make hydrogen produced via electrolysis from curtailed electricity from renewables cost-competitive even under low carbon prices.

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Introduction

Hydrogen can be a reliable source of energy provided a dependable supply at a reasonable price. Hydrogen is not found in nature and is classified as, grey (or brown), blue, and green depending on how it is produced. The two main methods to produce hydrogen are reformation, as with natural gas, and electrolysis. The former is still based on fossil fuels, making it not a sustainable solution. The latter can be sustainable if a reliable and environmentally friendly source of electricity is secured. Electricity generated from fossil fuels may not be desirable for electrolysis in the long term, but electricity generated from renewable sources could be desirable.

Hydrogen is expected to bring various benefits to the society. For instance, a transition to a hydrogen economy in the road transport sector in Germany is expected to bring environmental benefits through reducing greenhouse gas and air pollutant emissions and lead to make CO_2-free traffic possible in Germany [24]. Green hydrogen, hydrogen produced from renewables, can help an energy transition and link various energy sectors efficiently as a source of flexibility [18]. Specifically, green hydrogen that is produced from surplus hydroelectricity in Nepal appears to replace petroleum products in the transportation sector and lead to a sustainable energy solution [22]. However, against common understanding and expectation, green hydrogen led to an increase in net emissions and electricity price in Ireland. The marginal abatement cost of electrolysers appears to be 114.3€ per ton of CO_2 [13].

There are currently various methods of large-scale hydrogen production, namely steam reforming with CCS, alkaline electrolyser with stable or fluctuating power, lignite gasification and woody biomass generation [8]. Steam reforming with CCS presents the lowest cost followed by lignite gasification with CCS and woody biomass gasification. The cost of hydrogen from alkaline electrolyser with stable power is almost three times more expensive than that of steam reforming with CCS. Hydrogen from alkaline electrolyser with fluctuating power costs almost six times of that of steaming reforming with CCS. For the alkaline electrolyser, electrical feed cost is the main source of the high cost and followed by CAPEX and OPEX (OPEX excluding the feed cost) [8].

When electricity generated from renewable sources does not have corresponding demand, it causes negative net load. Net load is defined as the difference between forecasted load and expected electricity production from variable generation sources. Fig. 1 illustrates this net negative phenomenon with the load structure of a typical spring day in California. There are four typical ramping possibilities, namely at 4:00 a.m., 7:00 a.m., 4:00 p.m. and after sunset (or around 6:00 p.m.). The 4:00 p.m. possibility is the most significant daily ramp.

Curtailing renewables is one way of solving the oversupply problem, despite electricity being wasted [2]. To utilise this opportunity is using the electricity generated "otherwise curtailed" [9]. A study examined how curtailed renewables in Scotland can be used to produce green hydrogen. It showed that both reducing payments for the curtailment and producing green hydrogen are feasible [21].

One of the ways to mitigate oversupply is to increase demand; in other words, to offer consumers time-of-use rates, to increase energy storage, and to increase the ability of power plants to respond more quickly to changes in generation levels. The inherent variability of renewables such as wind or solar is an impediment to their effective use. Hydrogen fuel and electric power generation could be integrated at a wind farm. This allows flexibility of shifting production to best match resource availability given system operational needs and market factors. In times of excess electricity production from wind farms (i.e., the Belly of the Duck curve), instead of curtailing the electricity as is commonly done, this excess electricity can be utilised to produce hydrogen through electrolysis.

This study evaluates whether curtailed electricity from renewables, in the context of the Association of Southeast Asian Nations (ASEAN) and the East Asia Summit (EAS), can be used to produce hydrogen and how well it could serve these objectives in terms of economic and environmental perspectives. It estimates how much hydrogen is to be produced from curtailed electricity from variable renewables in the ASEAN and EAS regions.

The hydrogen produced would replace fossil fuels and, hence, reduce CO_2 emissions in the region. The amount of CO_2 emissions abated is treated as possible environmental benefits to be monetised by applying a prevailing carbon price in the US, the EU, or Singapore to the amount of emissions abated per unit of hydrogen produced from utilising curtailed electricity generated from renewables. This study surveys the cost of producing hydrogen through electrolysis in the literature and adopts a few estimates to examine the monetised environmental benefit against the cost of producing hydrogen via electrolysis.

There seems to be no study that applies a cost-benefit analysis to curtailed electricity for producing hydrogen against possible monetised benefits accrued from abated carbon dioxide emissions due to the produced hydrogen replacing fossil fuels. A few studies present the cost of producing hydrogen [8,17], but there are no extensions to compare the cost of producing hydrogen from curtailed electricity to possible monetised benefits from abated carbon dioxide emissions. Such a novel application of a cost-benefit analysis is to be the contribution of this study to the literature.

The rest of this study is structured as follows. Section Hydrogen: variable renewables and environmental implications reviews the types of hydrogen production, various cost estimates of hydrogen production from variable renewables, and environmental implications of hydrogen production from variable renewables via electrolysis. Section Methodology presents details of methodology applied in the study such as the cost estimation of producing hydrogen from curtailed electricity from variable renewables via electrolysis, the benefit estimation of producing hydrogen from curtailed electricity from renewables via electrolysis in the ASEAN and EAS regions, and estimates of carbon prices. Section Results, discussions, and policy implications discusses results and draws policy implications based on the results of the cost-benefit analysis conducted. Section Conclusions concludes this study.

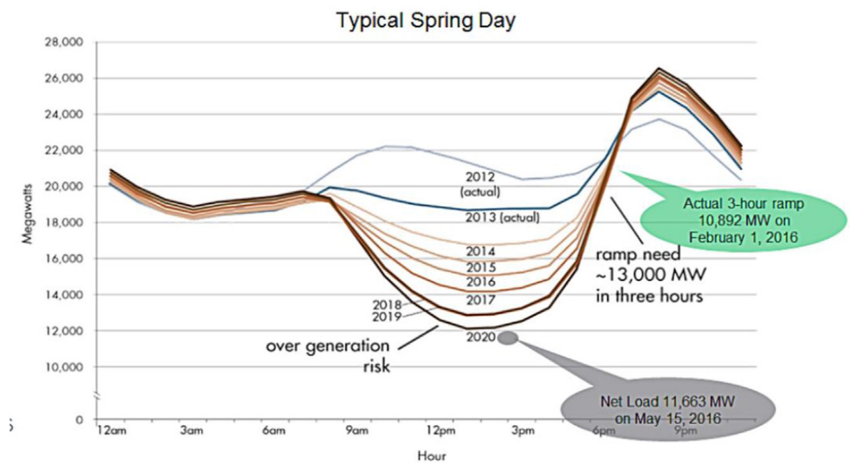

Fig. 1 – Load structure of a typical spring day in California, source: [2].

Hydrogen: variable renewables and environmental implications

Hydrogen and variable renewables

There are three types of hydrogen by production methods - grey or brown, blue and green hydrogen. Grey hydrogen is typically produced in a process called steam methane reformation, while brown hydrogen is produced from the coal (or ignite) gasification. Blue hydrogen is produced mainly from natural gas but with carbon capture and sequestration (CCS) technology. Green hydrogen is produced by a process called electrolysis in which a unit called an electrolyser that is powered splits water into hydrogen and oxygen using non-carbon-emitting electricity [5]. Electrolysers can assume various sizes, such as a small, appliance-size piece of equipment to a large-scale, central production facility. The former is good for small-scale distributed hydrogen production, while the latter is good for an electricity grid that is directly linked to renewable or non-carbon emitting electricity generation technology [23].

Apart from the types of hydrogen by production methods, there are various methods of large-scale hydrogen production, namely steam reforming with CCS, alkaline electrolysers with stable or fluctuating power, lignite gasification, and woody biomass generation [8]. Steam reforming with CCS presents the lowest cost, followed by lignite gasification with CCS and woody biomass gasification. The cost of alkaline electrolysers with stable power is almost three times that of steam reforming with CCS, while using alkaline electrolysers with fluctuating power costs almost six times that of steaming reforming with CCS. For the alkaline electrolyser, the electrical feed is the main source of the high cost, followed by capital and operation expenditure [8]. For making electrolysis a viable source of hydrogen, it is critical to reduce the capital cost of the electrolyser unit and the balance of the system, and to improve energy efficiency for converting electricity to hydrogen. Cost is the key determinant for the wide adoption of electrolysers. If these four criteria are met (i.e., low cost of power, low CAPEX, low BoP, and high efficiency), however, electrolysis could be a promising option for hydrogen production where renewable resources are used as the sources of electricity.

Hydrogen can play two key roles in the global economy, namely as a booster that accelerates more usage of renewable electricity and a channel that helps the global economy decarbonise. First, hydrogen can boost renewable energy by functioning as storage and a reliable source of cleaner energy. Second, hydrogen production via electrolysis can lead to zero carbon dioxide emissions depending on how electricity is generated, which could present a pathway to a net zero future. Green hydrogen has huge implications for storing energy generated from renewables such as wind and solar energy. Green hydrogen has been active in a few states in the US, such as California and Utah [14]. Considering the sheer volumes of decarbonising needed in the electricity sector, however, green hydrogen is not expected to be fully utilised in the near future. As a result, blue hydrogen must be fully developed and utilised before green hydrogen works at full scale [3].

Hydrogen has been cost-competitive and hailed as a low-carbon option. The Hydrogen Council pegs the 2020 cost of producing hydrogen via electrolysis from dedicated offshore wind in Europe to be on average USD6/kgH_2; it projects the cost to be USD2.6/kgH_2 in 2030. The breakdown of the drop in the cost is as follows: decreases in the capital cost (USD2.6), improvement in efficiency (USD0.4), decreases in the operations and maintenance cost (USD0.2), and decreases in the levelised cost of electricity of offshore wind (USD1.3) [7].

Table 1 presents the estimated costs of producing hydrogen by various sources in the E.U.

Compared to grey hydrogen, green hydrogen is still more expensive, as shown in Table 1. Green hydrogen in the EU is expected to cost-competitive by 2030. But it depends on not only the capital cost of electrolysers, but the electrolyser load factors along with the cost of electricity from renewable sources [1].

Hydrogen fuel can be integrated with electric power generation at the site of variable renewables, such as a wind farm. This allows production to be shifted to best match resource availability under system operational needs and market factors. When excess electricity is produced, it is possible to use it

Table 1 – Estimates of hydrogen production costs in the European Union.

Types of hydrogen	Costs (EUR/ kgH_2)	Costs (EUR/ MWh)
Gray (Fossil fuel hydrogen)	1.5	44.9
Blue (Low-carbon hydrogen)	2.0	59.9
Green (Clean hydrogen)	2.5–5.5	74.9–164.7

Source: [1,12].

to produce hydrogen through electrolysis rather than curtailing it [23].

The source of the required electricity has become critical in determining the benefits and economic viability of producing hydrogen via electrolysis. This includes the cost and efficiency of the electrolysis along with the amount of emissions coming from electricity generation. The current power grid of most countries is not suitable for providing the electricity needed to run the electrolysis sustainably because carbon dioxide and other greenhouse gases are released as long as fossil fuels are used. When hydrogen is produced via electrolysis with renewable sources from the grid or separate from the grid, such pathways will lead to virtually zero emissions of carbon dioxide and other greenhouse gases.

The pathway to hydrogen competitiveness requires developing more efficient "hydrogen" fossil fuel-based electricity production with CCS as a short-term goal and decreasing the cost of generating electricity from renewable sources as a long-term goal. Using curtailed electricity to produce hydrogen via electrolysis is a solution to meet the long-term goal.

The Hydrogen Council identified seven roles hydrogen can play in providing the sustainable pathway to the transformation of energy sector in the global economy. First, it enables large-scale renewable energy to be integrated in power generation. Second, it helps energy be distributed across sectors and regions. Third, it spurs energy system resilience. Fourth, it decarbonises transportation. Fifth, it decarbonises industrial energy use. Sixth, it promotes the decarbonisation of heat and power in the building sector. Seventh, it provides clean feedstock for industry [6]. Hydrogen works as an energy carrier and storage facility in energy transition [4]. Hydrogen is expected to be a transformative role in energy transition as a catalyst to decarbonise natural gas and accelerate uptake of CCS in producing blue hydrogen [5]. Solar photovoltaic and wind energy are expected to dominate in power generation and capture 62% of variable renewable share by 2050. Natural gas is expected to be the largest energy source in this decade and remain as the dominant source until 2050. These confirm the place of hydrogen in energy transition and the status of hydrogen in the energy system [5].

Hydrogen and environmental implications

Apart from helping the energy transition of the global economy, hydrogen is suggested as a key factor in the global economy to meet the temperature target set by the Paris Agreement [5]. Hydrogen can bring environmental benefits in providing a sustainable pathway to the transformation of energy sector in the global economy. First, hydrogen enables large-scale renewable energy to integrate in power generation so that it helps the global economy abate carbon dioxide emissions by displacing fossil fuels in power generation. Second, hydrogen helps decarbonises transportation, industrial energy use and heat and power in the building sector so that it displaces carbon-emitting fuels in sectors like transportation, industry, and the building sector [6].

The variable nature of renewable energy sources causes surplus in electricity supply at certain times and makes wasting the excess supply inevitable [2,9]. The production of hydrogen from curtailed electricity via electrolysis is expected to replace power generation from fossil fuels with renewable energy and hence reduce carbon dioxide emissions. Compared to the baseline case, the amount of carbon emissions in ASEAN countries can be reduced from 28% to 64% following the rates of fossil fuels replaced by using green hydrogen [16].

There are many studies on how hydrogen will help the global transition to a non-carbon emission economy with good cost estimates of hydrogen production via electrolysis. Along with cost studies, there are a few studies that present how hydrogen produced from variable renewables can help reduce carbon dioxide emissions. However, whether the expected reduction in carbon emissions by producing hydrogen via electrolysis from variable renewables is cost-effective against the accrued environmental benefits is not extensively studied.

Hydrogen produced from curtailed electricity from variable renewables abates carbon dioxide emissions, which can be monetised by a carbon price. By utilising a cost-benefit analysis, this study examined the cost-effectiveness of producing hydrogen from curtailed electricity from variable renewables via electrolysis. It takes the ASEAN hydrogen study [16] and extends it to the EAS region as a basis for estimating environmental benefits. Such an analytical framework can be applied to other regions provided a similar hydrogen production study exists. This would give at least some indication of the cost-competitiveness of producing hydrogen from curtailed electricity via electrolysis.

Methodology

The costs for using curtailed electricity from renewables for producing hydrogen are estimated using itemised cost inputs such as capitalisation expense (CAPEX), operating expense (OPEX), and feed cost, while the estimation of benefits is done by expected abatement of carbon dioxide emissions due to reduced usage of fossil fuels in the energy mix. The expected abatement of carbon dioxide emissions is monetised by applying current prices of carbon in China or the EU to the abated amount.

The cost-benefit analysis constructs a few scenarios with different costs of producing hydrogen via electrolysis in the market and discusses how carbon prices affect using the curtailed electricity surplus from renewables for producing hydrogen.

Cost estimates of producing hydrogen from curtailed electricity via electrolysis

There are various technical and economic factors in determining the cost of producing hydrogen via electrolysis. The electrolyser stack bears 50% and 60% of the CAPEX costs of alkaline and polymer electrolyte membranes, respectively. The power electronics, gas-conditioning, and plant components bear most of the rest of the costs. Apart from these factors, conversion efficiency, electricity costs, and annual OPEX are other key factors.

With more load hours of operation, the impact of CAPEX on the costs decreases but the electricity becomes the key cost component. For example, with CAPEX of USD450/kWe and a zero cost of electricity, the levelled cost of hydrogen could be about USD5/kgH_2 to less than USD1/kgH_2 over full load hours [8]. This illustrates how the curtailed electricity from renewable can be a competitive way of producing hydrogen.

Running the electrolyser at high full load hours and paying for the additional electricity can actually be cheaper than just relying on surplus electricity with low full load hours. The relationship between electricity costs and operating hours becomes apparent when looking at electrolysers that use grid electricity for hydrogen production [8]. Very low-cost electricity is generally available only for a few hours within a year, which implies a low utilisation of the electrolyser and high hydrogen costs that reflect CAPEX costs. With increasing hours, electricity costs increase, but the higher utilisation of the electrolyser leads to a decline in the cost of producing a unit of hydrogen up to an optimum level at around 3000 to 6000 equivalent full load hours. Beyond that, higher electricity prices during peak hours lead to an increase in hydrogen unit production costs. With all these considerations [8], asserts that ‘in the near term, hydrogen production from fossil fuels will remain the most cost-competitive option in most cases.’ This study aims to show producing hydrogen using curtailed electricity from renewables is a good alternative to fossil fuel-based hydrogen production.

A detailed economic assessment by the IEA, based on hourly solar and wind data over a year in five locations across different provinces, suggests hydrogen can be produced at a cost of USD 2–2.3/kgH2. In some provinces, the lowest production costs are reached by using only solar (Qinghai) or wind (Hebei and Fujian), while in Xinjiang and Tibet performance is best with a combination of the two [8].

[8] has provided various cost estimates for producing hydrogen by electrolysis. As this study uses curtailed electricity for producing hydrogen, it assumed the cost of electricity is zero.[3] The electrolyser load factor is the key determinant of the cost of producing hydrogen. If the electrolyser load factor is about 1500 h or higher, then the production cost could be less than USD2/kgH_2. If the electrolyser load factor is about 1000 h, then the cost could be USD2/kgH_2. If the electrolyser load factor becomes very low such as about 500 h, then the cost could be higher than USD10/kgH_2 [17].

This study takes two scenarios of cost estimates following two most plausible cases of the electrolyser load factor, namely 1000 h and 1500 h. The corresponding cost estimate is USD2/kgH_2 and USD1/kgH_2, respectively. Table 2 presents the cost estimates of producing hydrogen from curtailed electricity from variable renewables via electrolysis that this study adopts.

With more load hours of operation, the impact of CAPEX on the costs decreases but the electricity becomes the key cost component. This illustrates how the curtailed electricity from renewables can be the case of a competitive way of producing hydrogen. Running the electrolyser at high full load hours and paying for the additional electricity can actually be cheaper than just relying on surplus electricity with low full load hours.

Estimation of benefits from producing hydrogen from curtailed electricity in ASEAN

The power generation mix in ASEAN is estimated for Business as Usual (BAU) and various scenarios based on renewables-shares in the energy mix using the countries' energy models by applying the Long-range Energy Alternative Planning System (LEAP) software, an accounting system to project energy balance tables based on final energy consumption and energy input and/or output in the transformation sector. Final energy consumption is forecast using energy demand equations by energy and sector and future macroeconomic assumptions. The potential emission abatement is the difference between (a) the business as usual (BAU) scenario and (b) the alternative policy scenario (APS) and other high-renewable-share scenarios such as Senario1, Scenario2, and Scenario3.

The expected reduction of carbon dioxide emissions is monetised by current carbon prices in the US, the EU, or Singapore. This study takes hydrogen production from curtailed electricity in ASEAN and EAS region as the basis of calculating the possible amount of the emissions of carbon dioxide abated. Applying the LEAP software to ASEAN [16], presented various estimates of how much hydrogen can be produced from curtailed electricity.

In the modelling work applying LEAP, the baseline of 10 ASEAN member states was 2017, the real energy data available in 2017, which are the latest that the study employed. Projected demand growth is based on government policies, population, economic growth, and other key variables, such as energy prices used by the International Energy Agency energy demand model [8]. BAU is in line with current energy policy in the baseline information, which is used to predict future

Table 2 – Cost estimates of hydrogen production by electrolysis load factors.

Electrolyser load factors (hours)	Costs (USD/kgH_2)
1500	1
1000	2

Source: [8,17].

[3] The assumption that renewable generation cost will be zero (or at times negatives) is not realistic. The cost of export to the grid, maintenance costs, etc. will be paid by someone, so this can only happen under a legislative agreement led by Government. This study takes such an assumption for the simplicity of analysis. The cost estimates of the analysis constitute the lower bound of realistic costs.

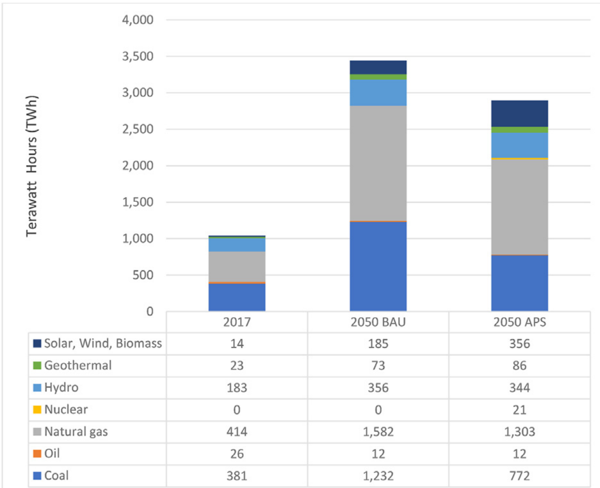

Fig. 2 – Power Generation Mix in ASEAN, BAU = business as usual, APS = Alternative Policy Scenario, Source: [16].

energy demand growth. However, APS differs from BAU in policy changes and targets, with a greater share of renewables, including possible nuclear uptake based on an alternative policy for energy sources and more efficient power generation and energy in final energy consumption.

Fig. 2 presents the power generation mix in ASEAN. This study also estimates how much hydrogen will be produced from curtailed electricity by extending the ASEAN study to the EAS region. Section Estimation of benefits from producing hydrogen from curtailed electricity in EAS. presents details of how much hydrogen will be produced from curtailed electricity in the EAS region.

The study constructs scenarios of how much renewables would replace total combined fossil fuel generation, i.e., by 10%, 20%, and 30% by 2050 and names them Scenario1 for 10% replacement, Scenario2 for 20% replacement and Scenario3 for 30% replacement. Each scenario further assumes the rate of curtailed electricity is utilised to produce hydrogen, namely at 20%–30%. Following these assumptions, the three scenarios are renamed Scenario1H_2, Scenario2H_2, and Scenario3H_2. Fig. 3 presents the share of combined fossil fuels in ASEAN, i.e. coal, oil, and natural gas, versus renewables under various scenarios.

The formulas to calculate potential hydrogen production in the renewable scenarios are as follows:

Scenario1H_2 (Mt-H_2) = [Scenario1 (TWh) x (Percentage of curtailed electricity) / 48 (TWh)]

Scenario2H_2 (Mt-H_2) = [Scenario2 (TWh) x (Percentage of curtailed electricity) / 48 (TWh)]

Scenario3H_2 (Mt-H_2) = [Scenario3 (TWh) x (Percentage of curtailed electricity) / 48 (TWh)]

Mt-H_2 stands for million tonnes of hydrogen; TWh is terawatt-hour; the percentage of curtailed electricity is 20% or 30% of total generation from renewables. The study also applies the conversion factor of 48 kW-hours (kWh) of electricity needed to produce 1 kg H_2 [10].

The potential emissions abatement is the difference between the BAU scenario and various renewable-share scenarios such as Scenario1, Scenario2, and Scenario3. Table 3 shows the amount of hydrogen produced from curtailed electricity by various replacement scenarios in ASEAN.

Table 4 presents the amount of carbon dioxide emissions that can be abated by different renewables replacement scenarios in 2050 in ASEAN. The amount of carbon dioxide emissions abated ranges from about 650 million tonnes to around 800 million tonnes.

Estimation of benefits from producing hydrogen from curtailed electricity in EAS

The potential of renewable hydrogen produced using curtailed electricity in Scenario1, Scenario2, and Scenario3 is quantified according to a renewable curtailment rate of 20%–30% for the high share of renewables in 2050 in EAS. Emissions abatement, i.e. the difference between BAU and various scenarios such as Scenario1, Scenario2, and Scenario3, is calculated. The higher share of renewables under Scenario1, Scenario2, and Scenario3 could only happen if hydrogen is developed as energy storage by utilising curtailed renewable electricity.

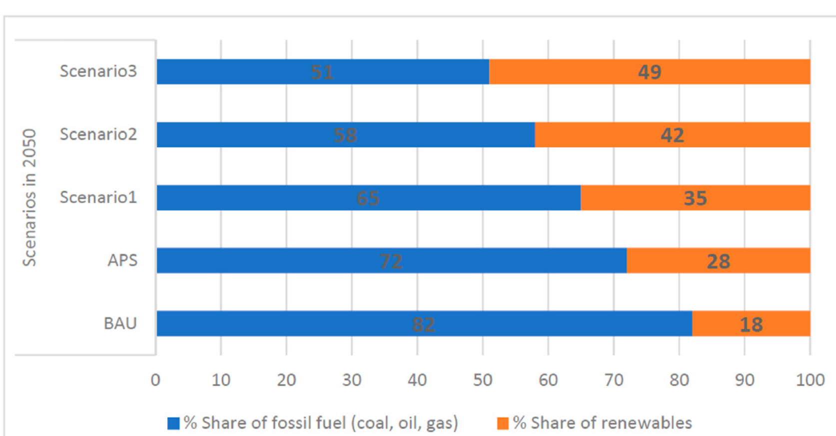

Fig. 3 – Share of Combined Fossil Fuels vs Renewables under Various Scenarios, BAU = business as usual, APS = Alternative Policy Scenario, Source: [16].

Table 3 – Hydrogen Produced from Curtailed Electricity in ASEAN (unit: million tonnes).

	Scenario1H_2	Scenario2H_2	Scenario3H_2
Of 20% curtailed renewables	4.23	5.10	5.97
Of 30% curtailed renewables	6.35	7.65	8.96

Source: [16].

Table 4 – Emissions Reduction by Scenarios in 2050 in ASEAN, (unit: million tonnes).

	Baseline (2017)	Emissions in 2050	Abatement in 2050
BAU	376	1216	–
Scenario1	376	568	648
Scenario2	376	506	710
Scenario3	376	442	774

BAU = business as usual.
Source: [16].

Table 5 – Hydrogen Produced from Curtailed Electricity in EAS (unit: million tonnes).

	Scenario1H_2	Scenario2H_2	Scenario3H_2
Of 20% curtailed renewables	72	76	80
Of 30% curtailed renewables	108	114	119

EAS = East Asia Summit.
Source: Authors' calculation (2021).

Utilising unused electricity and/or curtailed renewable electricity to produce hydrogen could be ideal to tap the maximum potential of renewables.

The higher share of renewables under various scenarios such as Scenario1, Scenario2, and Scenario3 will see a large reduction in carbon dioxide emissions, which could result in decarbonising emissions and contribute to the Conference of Parties (COP) commitments. The potential emissions abatement is the difference between the BAU scenario and various renewable-share scenarios such as Senario1, Scenario2, and Scenario3. Table 5 shows the amount of hydrogen produced from curtailed electricity by various replacement scenarios in EAS.

Table 6 presents the amount of the emissions of carbon dioxide can be abated by different renewable replacement scenarios in 2050 in EAS. Potential emissions abatement is 18,059 million tonnes-carbon (Mt-C), 18,527 Mt-C, and 18,994 Mt-C in Scenario1, Scenario2, and Scenario3, respectively.

Carbon prices

This study constructs a few cases with different hydrogen prices in the market. It adopts three carbon prices, namely, the Regional Greenhouse Gas Initiative (RGGI), European Union Emissions Trading System (EUETS) and a carbon tax in Singapore. For both RGGI and EUETS, the lowest and the highest carbon price are adopted. For the carbon price in Singapore, the current carbon tax is adopted (USD5 per tonne of CO_2). Table 7 presents various carbon prices used in the calculation of potential benefits of producing hydrogen from curtailed electricity via abating the emissions of carbon dioxide.

Table 6 – Emissions Reduction by Scenarios in 2050 in EAS (unit: million tonnes).

	Baseline (2017)	Emissions in 2050	Abatement in 2050
BAU	18,623	22,266	–
Scenario1	18,623	4207	18,059
Scenario2	18,623	3739	18,527
Scenario3	18,623	3272	18,994

BAU = business as usual.
Source: Authors' calculation.

Table 7 – Carbon prices (unit: USD/tonne-CO_2).

	Carbon price/Tax	Carbon price/Tax (USD)	Exchange rates
RGGI (the highest)	7.6	7.6	
RGGI (the lowest)	2.05	2.05	
EUETS (the highest)	49.88	60.15	@1.2059
EUETS (the lowest)	2.59	3.40	@1.3113
Singapore	5	3.77	@0.755

Source: [11,15,19,20].

Results, discussions, and policy implications

This study carries out a cost-benefit analysis of using curtailed electricity from renewables to produce hydrogen. The costs of this are compared to the possible monetised benefits accrued from the amount of the emissions of carbon dioxide abated due to reduced usage of fossil fuels in energy mix.

Results

ASEAN

Table 8 shows the amount of hydrogen produced from curtailed electricity under 20% curtailed renewables by different scenarios in the second row and the amount of the emissions carbon dioxide abated by different scenarios in the third row. The fourth row presents the amount of carbon dioxide abated per the unit amount of hydrogen produced from curtailed electricity from variable renewables via electrolysis. This is considered environmental benefits and the benefits are monetised by multiplying by carbon prices. In summary, Table 8 presents the possible amount of carbon dioxide abated per hydrogen produced in 2050.

Table 8 – Possible Amounts of CO_2 Abated per H_2 Production in 2050 (unit: million tonnes).

	Scenario1H_2	Scenario2H_2	Scenario3H_2
H_2 production in 2050	4.23	5.10	5.97
CO_2 abated in 2050	648	710	734
CO_2 abated per H_2 produced	153.19	139.22	122.95

Source: [16].

Using the carbon price presented in Table 7 and the possible amount of carbon dioxide abated per hydrogen produced (Table 8), the possible monetised benefits of unit of hydrogen produced from curtailed electricity from variable renewables via electrolysis is estimated. Table 9 shows the possible monetised benefits by different carbon prices expressed in US dollars per kgH_2, which accrue from hydrogen that replaces fossil fuels in the energy mix in ASEAN.

EAS

Table 10 shows the amount of hydrogen produced from curtailed electricity under 20% curtailed renewables by different scenarios in the second row and the amount of the emissions carbon dioxide by different scenarios in the third row. The fourth row presents the amount of the carbon dioxide abated per the unit amount of hydrogen produced from curtailed electricity from variable renewables via electrolysis. In sum, Table 10 presents the possible amount of CO_2 abated per hydrogen produced in 2050 in EAS.

Using the carbon price presented in Table 7 and the possible amount of CO_2 abated per hydrogen produced (Table 10), the possible monetised benefits per unit of hydrogen produced from curtailed electricity from variable renewables via electrolysis are estimated. Table 11 shows the possible monetised benefits by different carbon prices expressed in USD per kgH_2, which accrue from hydrogen that replaces fossil fuels in energy mix in EAS.

Discussions

As noted in Section Methodology, the electrolyser load factor is the key determinant of the cost estimate of producing hydrogen from curtailed electricity. The production cost is about USD2/kgH_2 when the electrolyser load factor is 1000 h. Except the case of the highest carbon price in EUETS, the cost appears to be higher than the benefit for all other cases. With the higher electrolyser load factor of 1500 h, the cost-benefit ratio could be lower than one for the RGGI highest carbon price.

Table 9 – Possible benefits of hydrogen produced from curtailed electricity (unit: USD/kgH_2).

	Scenario1H_2	Scenario2H_2	Scenario3H_2
RGGI (the highest)	1.16	1.06	0.93
RGGI (the lowest)	0.31	0.29	0.25
EUETS (the highest)	9.21	8.37	7.40
EUETS (the lowest)	0.52	0.47	0.42
Singapore	0.58	0.53	0.46

Source: Authors' own calculations.

Table 10 – Possible Amounts of CO_2 Abated per H_2 Production in 2050 (unit: million tonnes).

	Scenario1H_2	Scenario2H_2	Scenario3H_2
H_2 production in 2050	72	76	80
CO_2 abated in 2050	18,059	18,527	18,994
CO_2 abated per H_2 produced	250.82	243.78	237.43

Source: Authors' own calculation.

Table 11 – Possible benefits of hydrogen produced from curtailed electricity (unit: USD/kgH_2).

	Scenario1H_2	Scenario2H_2	Scenario3H_2
RGGI (the highest)	1.91	1.85	1.80
RGGI (the lowest)	0.51	0.50	0.49
EUETS (the highest)	15.09	14.66	14.28
EUETS (the lowest)	0.85	0.83	0.81
Singapore	0.95	0.92	0.90

Source: Authors' own calculations.

The cost-benefit ratio is calculated by the two cost estimates (Table 2) over the possible monetised benefits (Tables 9 and 11 for ASEAN and EAS, respectively). Table 12 presents cost-benefit ratios under the electrolyser load factor is 1500 h in ASEAN. Two scenarios of RGGI (the highest carbon price) and EUETS (the highest carbon price) show the cost-benefit ratio is lower than one, which implies those options are cost-effective and plausible.

Table 13 presents cost-benefit ratios under the electrolyser load factor is 1000 h in ASEAN. Only EUETS (the highest carbon price) shows a cost-benefit ratio of lower than one, which indicates the options are cost effective.

Table 14 presents cost-benefit ratios under the electrolyser load factor being 1500 h in EAS. Two scenarios of RGGI (the

Table 12 – Cost-benefit ratios under the electrolyser load factor (1500 h) in ASEAN.

	Scenario1H_2	Scenario2H_2	Scenario3H_2
RGGI (the highest)	0.86	0.95	1.07
RGGI (the lowest)	3.18	3.50	3.97
EUETS (the highest)	0.11	0.12	0.14
EUETS (the lowest)	1.92	2.11	2.39
Singapore	1.73	1.90	2.15

Source: Authors' own calculations.

Table 13 – Cost-benefit ratios under the electrolyser load factor (1000 h) in ASEAN.

	Scenario1H_2	Scenario2H_2	Scenario3H_2
RGGI (the highest)	1.72	1.89	2.14
RGGI (the lowest)	6.37	7.01	7.94
EUETS (the highest)	0.22	0.24	0.27
EUETS (the lowest)	3.84	4.23	4.79
Singapore	3.46	3.81	4.31

Source: Authors' own calculations.

Table 14 – Cost-benefit ratio under the electrolyser load factor (1500 h) in EAS.

	Scenario1H_2	Scenario2H_2	Scenario3H_2
RGGI (the highest)	0.52	0.54	0.55
RGGI (the lowest)	1.94	2.00	2.05
EUETS (the highest)	0.07	0.07	0.07
EUETS (the lowest)	1.17	1.21	1.24
Singapore	1.06	1.09	1.12

Source: Authors' own calculations.

Table 15 – Cost-benefit ratio under the electrolyser load factor (1000 h) in EAS.

	Scenario1H_2	Scenario2H_2	Scenario3H_2
RGGI (the highest)	1.05	1.08	1.11
RGGI (the lowest)	3.89	4.00	4.11
EUETS (the highest)	0.13	0.14	0.14
EUETS (the lowest)	2.35	2.42	2.48
Singapore	2.11	2.17	2.23

Source: Authors' own calculations.

highest carbon price) and EUETS (the highest carbon price) show the cost-benefit ratios are lower than 1, which implies those options are cost-effective and plausible.

Table 15 presents cost-benefit ratios under the electrolyser load factor is 1000 h in EAS. Only EUETS (the highest carbon price) shows the cost-benefit ratio is lower than one, which indicates the options are cost effective.

Unlike ASEAN, the cost-benefit analysis for EAS shows a ratio of close to 1 under the highest carbon price of RGGI. The larger production of hydrogen from curtailed electricity from variable renewables via electrolysis replaces larger amount of fossil fuels in energy mix in EAS. This leads to the larger amount of CO_2 emissions abated from which larger monetised benefits are accrued. Larger potential in hydrogen production in EAS appears to make the cost-benefit ratio close to 1 even under the domain of a low carbon price such as USD10 per ton of CO_2.

The current carbon prices in RGGI appear not to make producing hydrogen from curtailed electricity cost-effective or plausible. The carbon prices need to double the current price levels if the electrolyser load factor is about 1000 h. If the electrolyser load factor is about 1500 h or higher, then the current carbon prices seem to be cost-effective for producing hydrogen via electrolysis from curtailed electricity.

The carbon prices in EUETS have been above EUR20 per tonne of CO_2 since 2020. The current carbon prices in EUETS appear to make producing hydrogen from curtailed electricity cost-effective or plausible. If such trends in carbon prices in EUETS continue, then producing hydrogen via electrolysis from curtailed electricity seems to be cost effective or plausible in European nations.

The current carbon price in Singapore will not make producing hydrogen via electrolysis from curtailed electricity plausible. Carbon prices should be twice higher than the current carbon tax for the electrolyser load factor of 1500 h and more than quadruple the current carbon tax for the electrolyser load factor of 1000 h.

[17] presents the cost could be higher than 10USD/kgH_2 if the electrolyser load factor is 500 h or lower. A low carbon price would make the monetised environmental benefits accrued from abated CO_2 emissions low, which in turn makes hydrogen production from curtailed electricity from variable renewables via electrolysis less attractive if not attractive at all.

In sum, producing hydrogen from curtailed electricity from variable renewables via electrolysis could be a plausible option even under low carbon price if the electrolyser load factor is higher than 1500 h. If carbon prices are sufficiently high such as higher than USD20/tonne of CO_2, then producing hydrogen from curtailed electricity from variable renewables via electrolysis could be cost effective even under the low level of the electrolyser load factor such as 1000 h. If the electrolyser load factor is very low such as 500 h, then curtailed electricity from variable renewables could be a plausible option for producing hydrogen only if the carbon prices are significantly high, e.g., in the range of USD 80 to USD 100/tonne of CO_2.

Policy implications

The cost-benefit analyses of producing hydrogen from curtailed electricity present the plausible range of carbon prices that make curtailed electricity a cost-effective source of producing hydrogen. This study derives the following policy implications.

First, it is good to grant the abated amount of CO_2 to get carbon credits so that they can be traded in the carbon market. A regional carbon market is expected to promote the utilisation of hydrogen production from curtailed electricity from variable renewables. The hydrogen produced works as an energy storage that would solve the intrinsic intermittency of variable renewables and excess supply of electricity from variable renewables.

Second, the cost of producing hydrogen via electrolysis varies by the electrolyser load factors. It is beneficial to make the electrolyser load factors 1000 h or above. With this, producing hydrogen from curtailed electricity from renewables is viable even low carbon prices.

Conclusions

This study examines if curtailed electricity can be a cost effective and plausible option for producing hydrogen in the Association of Southeast Asian Nations (ASEAN) and East Asia Summit (EAS) region employing a cost-benefit analysis. The cost estimation is based on two scenarios of the electrolyser load factor, namely 1000 h and 1500 h. The electrolyser load factor appears to be the key determinant of the cost estimate of producing hydrogen from curtailed electricity from renewables via electrolysis. The production cost is about USD2/kgH_2 when the electrolyser load factor is 1000 h. The benefit of producing hydrogen from curtailed electricity from renewables via electrolysis is estimated by three stages. At a first stage, the amount of fossil fuels to be replaced by the amount of hydrogen produced from curtailed electricity is estimated. At a second stage, the amount of CO_2 emissions to be abated by the reduced usage of fossil fuels in energy mix is estimated. At a third and final stage, the amount of CO_2 emissions abated by the unit amount of hydrogen produced is estimated, which is monetised by multiplying a prevailing carbon price.

The cost of producing hydrogen from curtailed electricity from variable renewables via electrolysis appears to be higher than the benefit accrued from the abated amount of CO_2 emissions for all but the case of the highest carbon price of European Union Emissions Trading System (EUETS) for ASEAN and EAS. With the higher electrolyser load factor of 1500 h, the cost-benefit ratio could be lower than one for the highest carbon price of the Regional Greenhouse Gas Initiative (RGGI) for EAS.

The current carbon prices of RGGI or Singapore appear not to make producing hydrogen from curtailed electricity cost-effective or plausible. The carbon prices of EUETS, which are above EUR20 per tonne of CO_2, appear to make producing hydrogen from curtailed electricity from variable renewables via electrolysis cost effective or plausible. The current carbon price in Singapore appears not to make producing hydrogen from curtailed electricity from variable renewables via electrolysis plausible.

In sum, producing hydrogen from curtailed electricity from renewables via electrolysis could be a plausible option under two occasions, either high carbon prices or the electrolyser load factor is higher than 1500 h. If carbon prices are significantly high such as higher than 15 USD/ton of CO_2, then producing hydrogen from curtailed electricity from renewable via electrolysis could be cost effective even under the low level of the electrolyser load factor such as 1000 h. If the electrolyser load factor is very low such as 500 h, then producing hydrogen from curtailed electricity from renewables could be a plausible option for producing hydrogen only if the carbon prices are significantly high such as in the range of USD80 to USD100/tonne of CO_2.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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